PEDDIREDDY SURYAPRAKASH

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SUMMARY

Mining Engineering graduate from IIT Kharagpur transitioning toward Computational Science and High-Performance Computing. Strong foundation in C++ programming, algorithms, numerical methods, linear algebra, and performance-oriented computing. Built C++17/OpenMP projects including a parallel Jacobi solver and 2D stencil engine to study parallel speedups, cache behavior, and memory bandwidth limits. Interested in scientific computing, parallel programming, numerical linear algebra, performance optimization, and AI systems infrastructure.

EDUCATION

Indian Institute of Technology (IIT) Kharagpur | West Bengal, India | 07/2025
Bachelor of Technology (B.Tech) : Mining Engineering

Cumulative GPA**:** 7.04 / 10.00
Relevant Coursework and Training:

- **Computational & Programming:** Programming and Data Structures (Theory & Laboratory), Computational Methods and Mine Instrumentation, Mine Computation Laboratory
- **Mathematics & Statistics:** Advanced Calculus, Linear Algebra, Numerical and Complex Analysis , Probability and Statistics
- **Management & Economics:** Quantitative Decision Making, Mineral Economics and Business

Bhashyam High School | Guntur, India | 05/2020
Higher Secondary Certificate (HSC) — BIEAP

- Cumulative GPA: 9.94/10.00

PROJECTS

Parallel Jacobi Linear System Solver | C++17, OpenMP, Multi-Threading | GitHub

- Implemented an OpenMP-parallelized Jacobi iterative solver in C++17 for 2500 × 2500 linear systems, achieving a 4.81× speedup on 8 CPU cores.
- Designed a thread-safe iterative loop using OpenMP reduction(max) primitives to track residual error convergence without mutex locks.
- Generated strictly diagonally dominant matrices to support numerical stability and convergence across multi-threaded runs.
- Structured the implementation across modular C++ header and source files and compiled with GCC -O3 optimization flags.
- Strengthened understanding of numerical linear algebra, convergence behavior, thread-level parallelism, and performance benchmarking.

Parallel 2D Spatial Image Stencil Engine | C++17, OpenMP, Cache Optimization | GitHub

- Developed a 2D spatial averaging stencil engine in C++17 operating on a 16-megapixel 4000 × 4000 grid over 50 iterations.
- Implemented row-major memory layout using a flat 1D contiguous vector to improve cache locality during stencil operations.
- Built automated timing harnesses using omp_get_wtime() to measure runtime, speedup, and parallel efficiency across 1, 2, 4, and 8 threads.
- Analyzed low arithmetic intensity and memory bandwidth limitations, identifying the “memory wall” effect where speedup plateaued due to memory contention.
- Gained practical exposure to shared-memory parallelism, cache-aware programming, and performance bottleneck analysis.

EXPERIENCE

Web Development Intern |Unified Mentor Pvt Ltd | Gurgaon, India | 06/2024 - 07/2024

- Worked on backend workflow components for structured user data handling, validation, and persistence.
- Assisted in organizing database schemas and server-side logic to support efficient request handling.
- Contributed to integration between application layers, focusing on clean data flow and input validation.
- Gained exposure to backend development practices, database interaction, and software component integration.

Intern|The Singareni Collieries Company Limited (SCCL) | Telangana, India | 05/2024 - 06/2024

- Completed a structured field engineering internship at the Ravindrakhani No. 6 Incline underground coal operations.
- Observed daily underground Board and Pillar face operations, continuous miner setups, and mechanized material handling workflows.

SKILLS

- Programming Languages: C++17, Python, OpenMP
- Parallel Computing & Systems: Shared-memory multithreading, OpenMP parallelization, cache-aware programming, row-major memory layouts, SIMD/vectorization awareness, performance benchmarking
- Algorithms & Mathematics: Data Structures and Algorithms, Graph Theory, Dynamic Programming, Linear Algebra, Numerical Analysis, Calculus, Probability and Statistics
- Tools & Environments: Git, GitHub, GCC/G++, Linux/Unix CLI, PowerShell
- Independent Technical Preparation: C++ systems programming, scientific computing fundamentals, numerical linear algebra, parallel programming, performance analysis

ACADEMIC / CAREER PREPARATION PERIOD

UPSC Civil Services Examination Preparation | India | 2025 – 2026

- Prepared full-time for the UPSC Civil Services Examination while independently strengthening programming, numerical computing, and parallel computing fundamentals.
- Used this period to clarify long-term academic direction toward Computational Science, High-Performance Computing, and systems-oriented software engineering.

ACCOMPLISHMENTS

- JEE Advanced: All India Rank 8,000
- **Competitive Programming:** LeetCode peak rating of 1874; solved 1,500+ algorithmic problems across LeetCode